

T5 Task Analogy Ablations Improvement

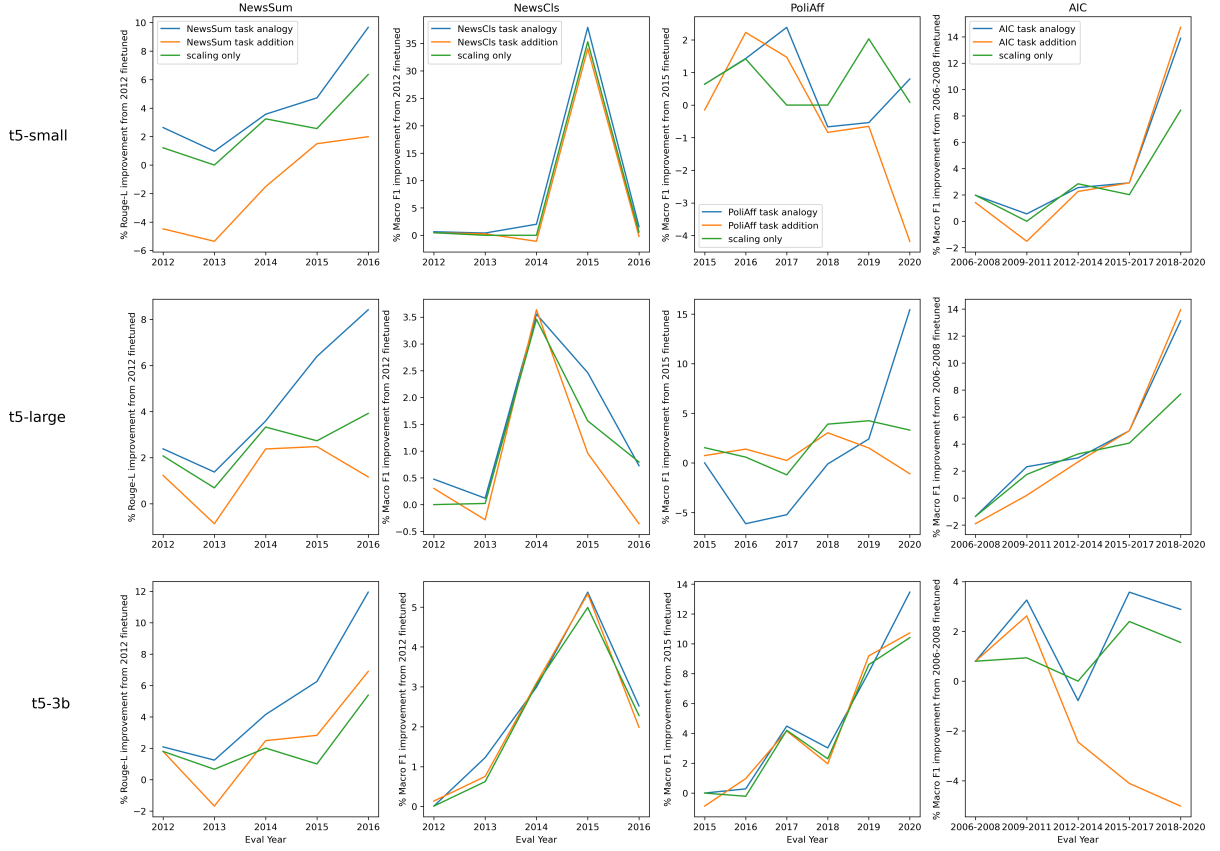


Figure 15: Time vector analogy ablations for three sizes of T5. Given the time vector analogy $\tau_k \approx \alpha_1 \cdot \tau_j + (\alpha_2 \cdot \tau_k^{\text{LM}} - \alpha_3 \cdot \tau_j^{\text{LM}})$, $\alpha_1, \alpha_2, \alpha_3 \neq 0$, we define "task addition" to be only adding the language modeling vector (i.e. $\alpha_1, \alpha_2 \neq 0, \alpha_3 = 0$), and "scaling only" to be only scaling the base τ_j model (i.e. $\alpha_1 \neq 0, \alpha_2, \alpha_3 = 0$). We sweep over the same alpha combinations as in §4.4 and report the best results for each target year, task, and T5-size.

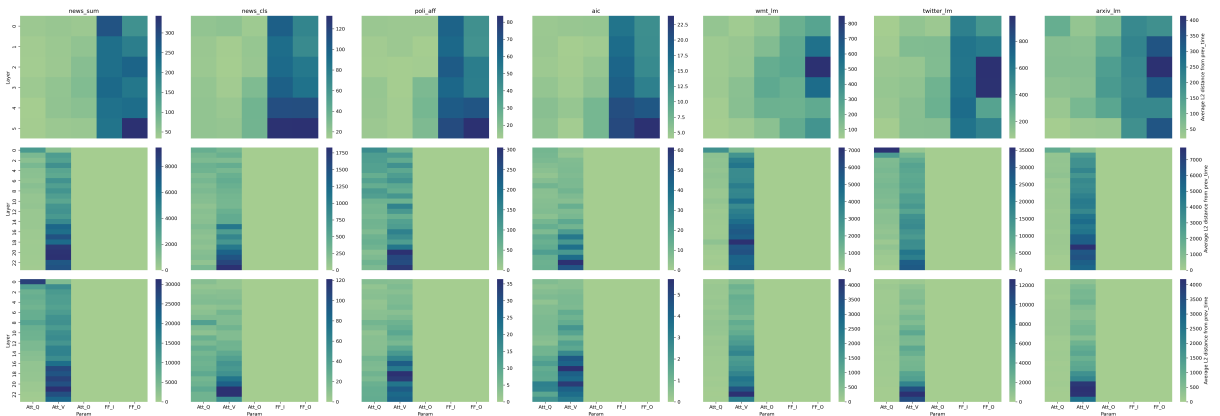


Figure 16: Year-to-year, T5-small feed forward layers change the most across all tasks and domains, and attention changes more in the language modeling setting. For our T5-large and T5-3b models trained with LoRA, the V attention layers change more than the Q layers, with most of the changes (regardless of model size) concentrated in the last layers. Like our param swapping experiment, this suggests that some parameters play a larger role in temporal misalignment than others.